SPATIAL ANALYSIS OF MULTI-HAZARD RISK AND ACCESSIBILITY OF CRITICAL FACILITIES IN MARIKINA CITY

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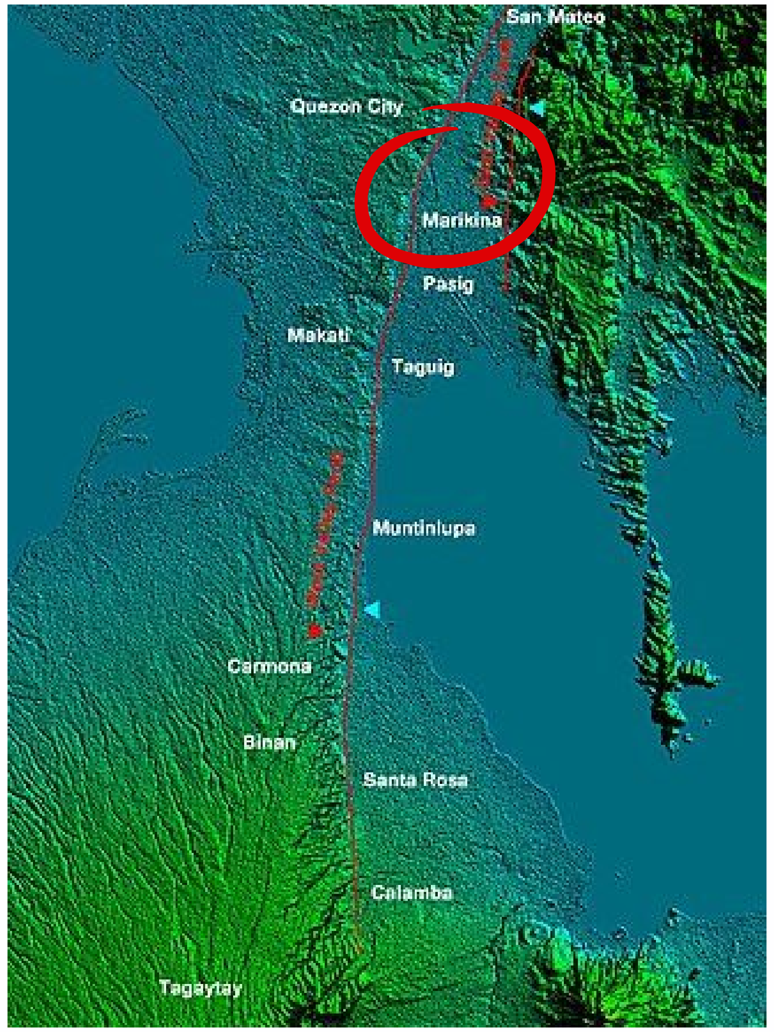
TO RECAP...

Marikina City is one of Metro Manila’s most **flood-prone** areas and is situated along the **West Valley Fault**, making it exposed to multi-hazard risk areas in the region.



OBJECTIVES

- To map the spatial distribution of critical facilities in Marikina City alongside flood hazard zones and fault line data

- To determine which critical facilities are located in close proximity to the West Valley Fault

- To identify which critical facilities and barangays in Marikina City are most likely to be exposed to flood hazards

RESEARCH QUESTIONS

1. Which critical facilities in Marikina City are located within flood hazard zones?
2. What critical facilities in Marikina City are situated near active fault lines?
3. Which barangays in Marikina City are most exposed to flood and seismic activity?

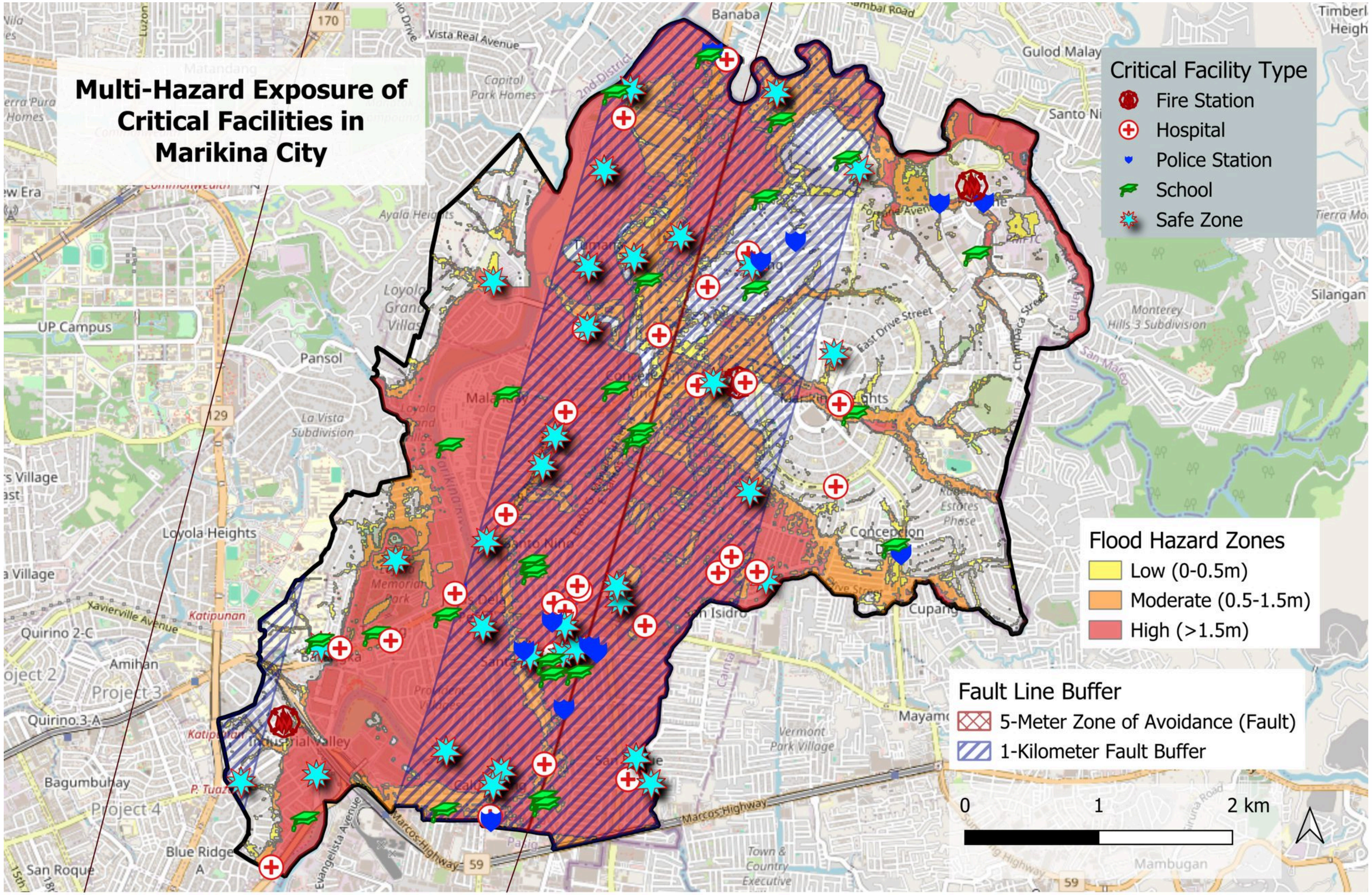
DATA AND SOURCES

Dataset	File Format	Source	Description
NOAH Hazard Maps	ESRI Shapefile	UP Resilience Institute (UPRI)	Spatial layers for Flood (100-yr Return)
Fault Line Data	ArcGIS Feature Layer	Philippines Institute of Volcanology and Seismology (DOST-PHIVOLCS)	Distribution of Active Faults in the Philippines
Critical Facilities	ESRI Shapefile	OpenStreetMap (OSM)	Point locations for Schools, Hospitals, Fire Stations, and Police Stations.

DATA AND SOURCES

Dataset	File Format	Source	Description
Barangay Boundaries	ESRI Shapefile	National Mapping and Resource Information Authority (NAMRIA), Philippines Statistics Authority (PSA)	Administrative Boundaries for spatial aggregation
2020 Census of Population and Housing	Excel Workbook File	Philippines Statistics Authority (PSA)	Population count based on region and/or barangay
Safe Zones in Marikina City	ESRI Shapefile	Manila Bulletin	Designated safe zone and evacuation areas in Marikina City (manually inputted as supplementary data

MAP I



Data Sources: NOAH Flood Hazard Maps (UPRI) | Active Fault Line (PHIVOLCS via ArcGIS Online) | Critical Facilities (OpenStreetMap) | Safe Zones (Marikina LGU via Manila Bulletin) | Administrative Boundaries (NAMRIA/PSA)

Map Design by: Meia Go, Rovin Montano

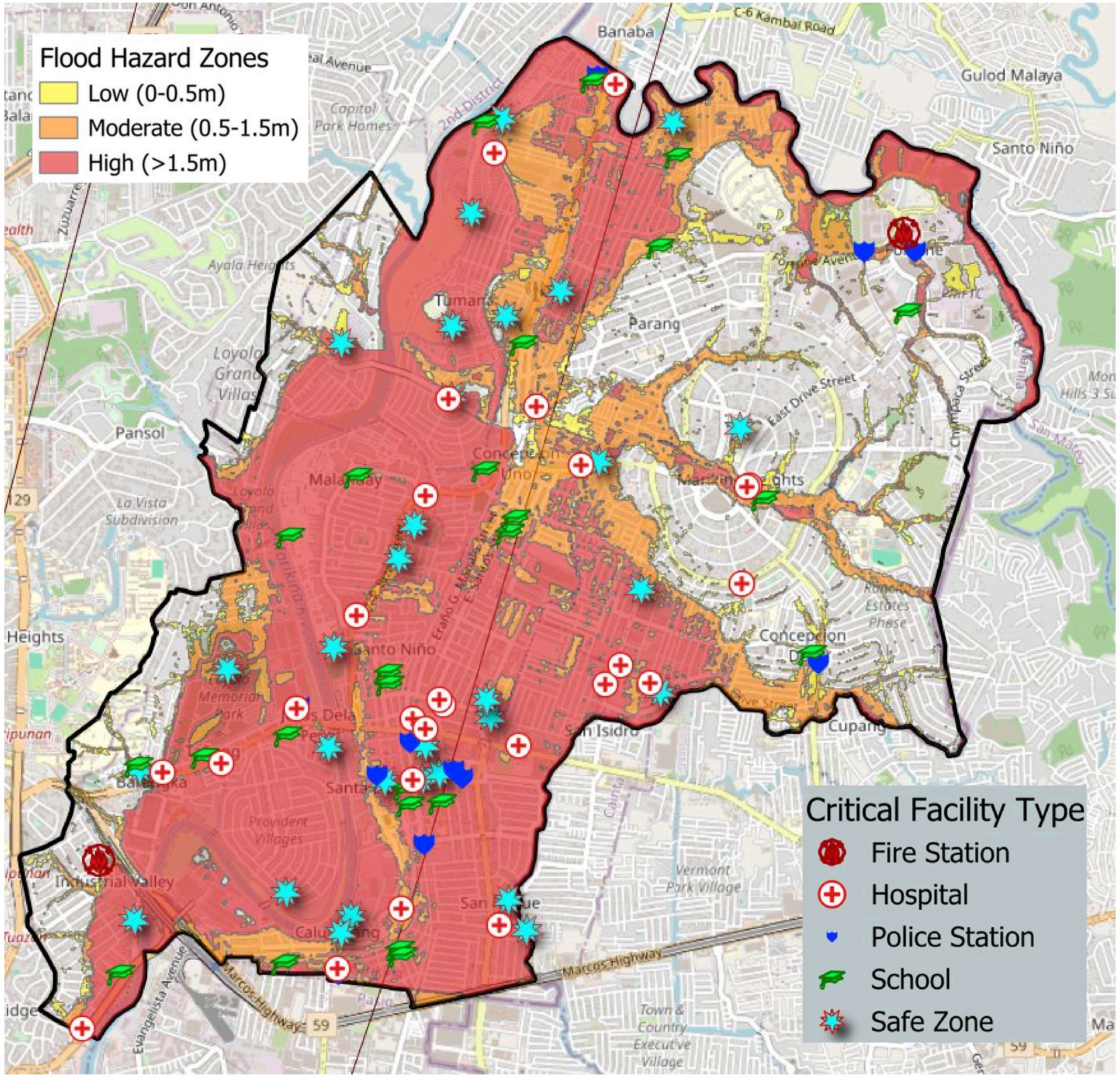
RESEARCH QUESTIONS

1

Which critical facilities in Marikina City are located within flood hazard zones?

Critical Facility	High Hazard	Moderate	Low Hazard
Hospital	19	6	1
Fire Stations	1	1	N/A
Schools	19	3	3
Police Stations	8	3	N/A
Safe Zones	25	3	2
Total:	**72**	**16**	**6**

RESEARCH QUESTIONS



Flood Hazard Zones in Marikina City

RESEARCH QUESTIONS

2

What critical facilities in Marikina City are situated near active fault lines?

Within the 5-meter Zone of Avoidance:

NONE

RESEARCH QUESTIONS

2

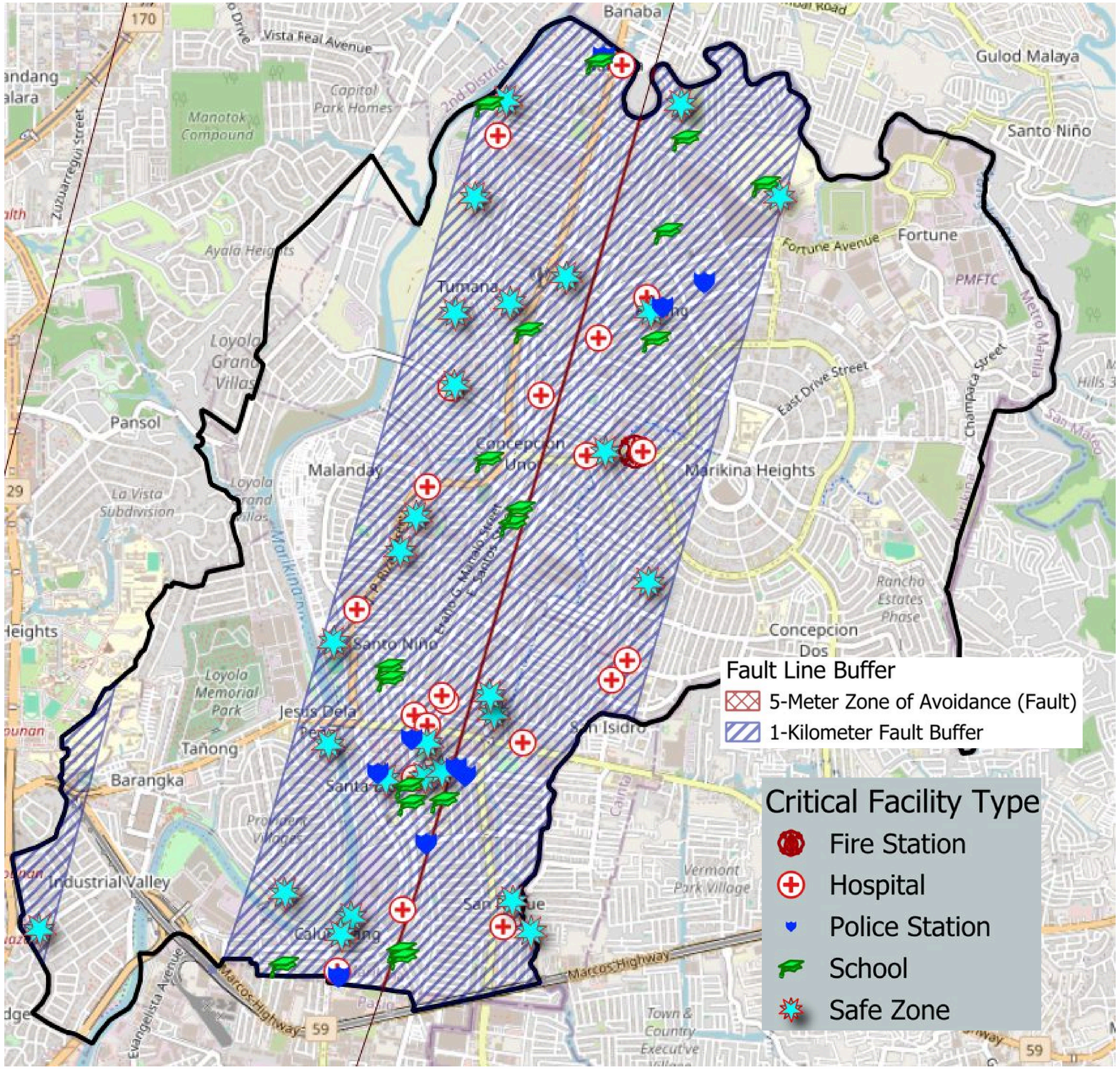
What critical facilities in Marikina City are situated near active fault lines?

Within a 1-kilometer buffer zone:

76

Critical Facility Type	No. of Facilities Affected
Hospital	21
Fire Stations	1
Schools	18
Police Stations	9
Safe Zones	27
Total:	76

RESEARCH QUESTIONS



Facilities that Fall Within 5m and 1km Fault Line Buffers in Marikina City

ADDITIONAL INFORMATION

What critical facilities in Marikina City are exposed to both hazards (high flood hazard zone and within 1-km fault line buffer)?

Critical Facility Type	No. of Facilities Affected
Hospital	14
Fire Stations	0
Schools	14
Police Stations	6
Safe Zones	21
Total:	55

Total Critical Facilities Exposed:

55

ADDITIONAL INFORMATION

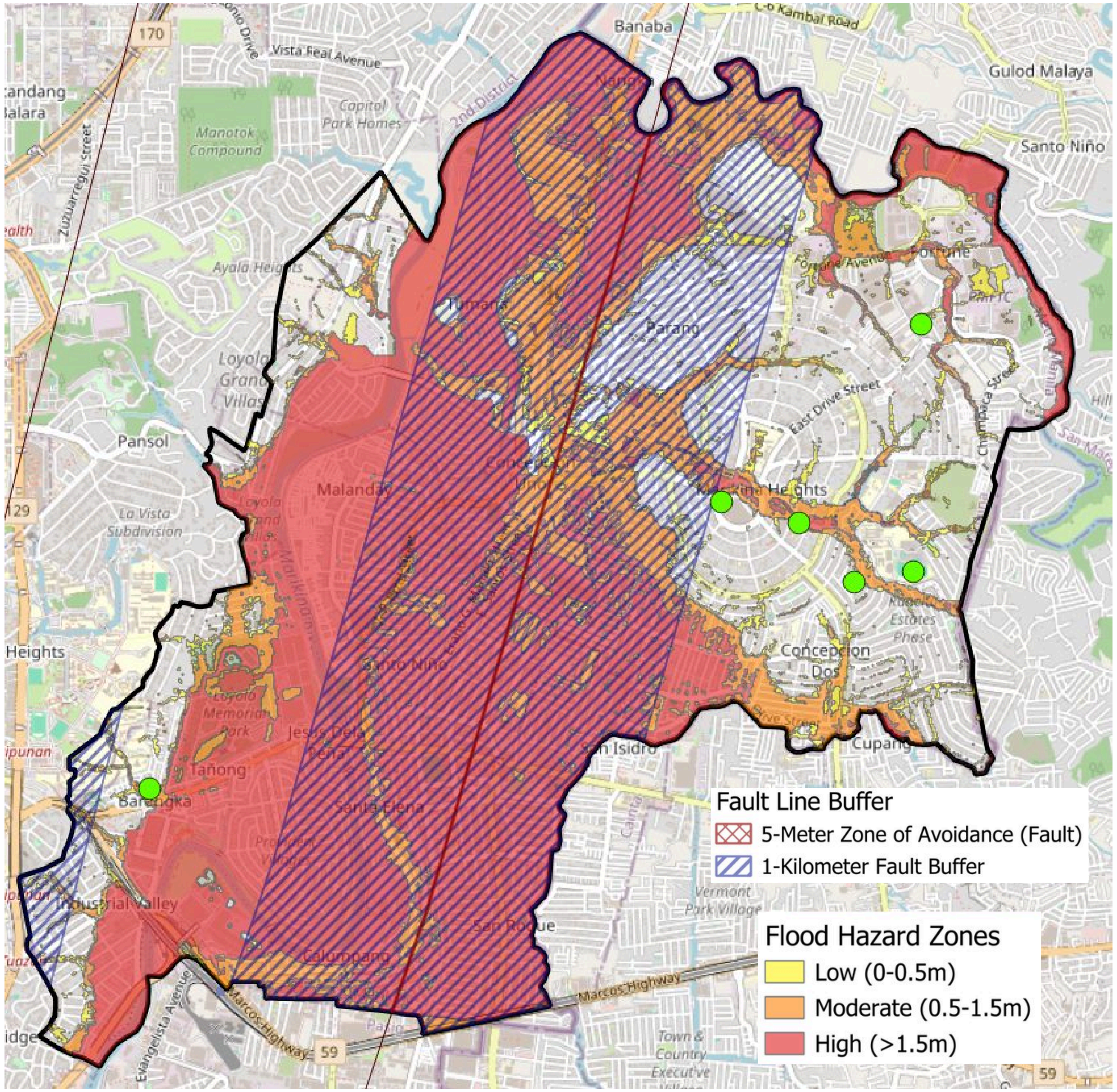
What critical facilities in Marikina City are not exposed to either hazards (based on the flood return data and 1-km fault line buffer)?

Name	Critical Facility Type
Rancho Estates Open Space	Safe Zone
Liwasang Kalayaan	Safe Zone
Marist Open Space	Safe Zone
Barangka Elementary School	School
Marikina Heights National High School	School
Fortune Elementary School	School

Total Critical Facilities:

6

RESEARCH QUESTIONS



Critical Facilities Not Within the Fault Line Buffers and Not Within Flood Hazard Zones

Represented by:

MAP 2 METHODOLOGY

Data Used:

- Hazard exposure results from Map 1
- Barangay boundaries (NAMRIA)
- 2020 Census of Population and Housing (PSA)

Step-by-step:

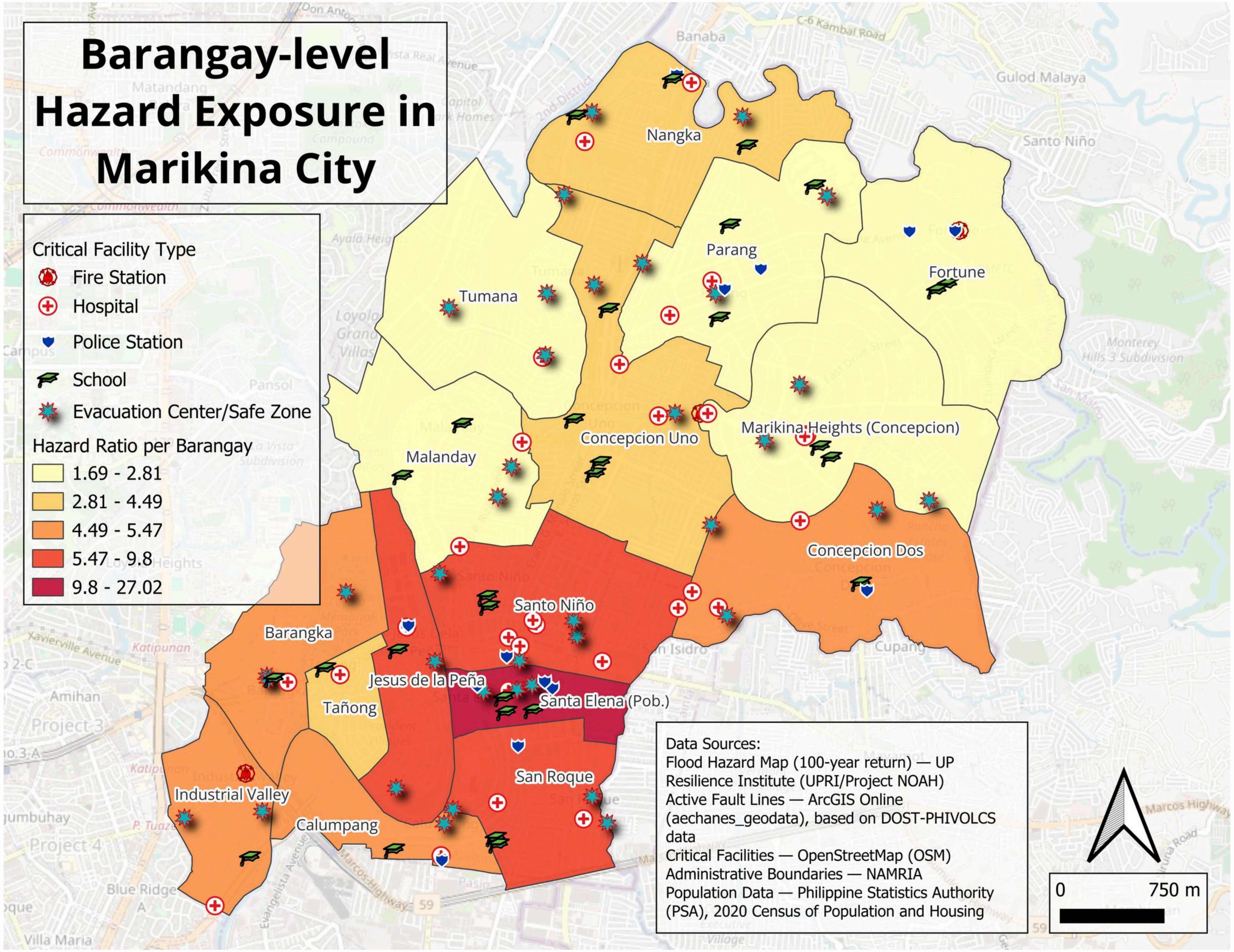
1. Assign exposure scores to each facility from Map 1 results
    - 2 = exposed to both flood and seismic hazards
    - 1 = exposed to one hazard
    - 0 = not exposed
2. Aggregate scores per barangay using Join Attributes by Location (Summary)
3. Join PSA 2020 population data to the barangay layer
4. **Compute hazard ratio using the Field Calculator:**

Hazard Ratio = (Exposure Score / Barangay Population Total) × 10,000

Interpreted as: weighted hazard exposure per 10,000 residents

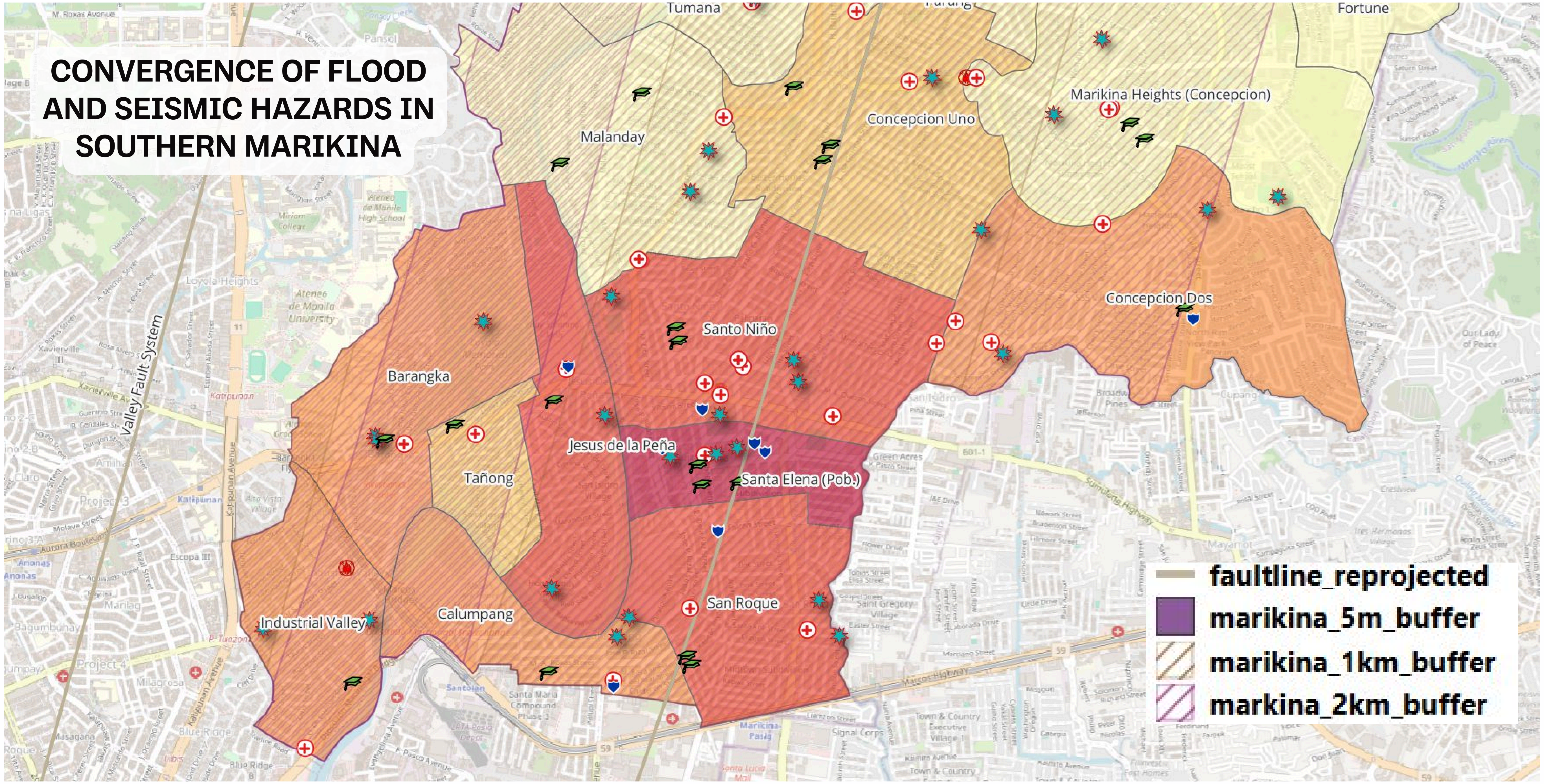
5. Apply graduated symbology (Natural Breaks / Jenks, 5 classes, YlOrRd color ramp)
6. Overlay critical facility points by type

MAP 2



Barangay-level Hazard Exposure in Marikina City
Critical Facility Type
Fire Station
Hospital
Police Station
School
Evacuation Center/Safe Zone
Hazard Ratio per Barangay
1.69 - 2.81
2.81 - 4.49
4.49 - 5.47
5.47 - 9.8
9.8 - 27.02
Nangka
Parang
Fortune
Tumana
Concepcion Uno
Marikina Heights (Concepcion)
Malanday
Concepcion Dos
Santo Niño
Barangka
Jesus de la Peña
Tañong
Santa Elena (Pob.)
San Roque
Industrial Valley
Calumpang
Data Sources:
Flood Hazard Map (100-year return) — UP Resilience Institute (UPRI/Project NOAH)
Active Fault Lines — ArcGIS Online (aechanes_geodata), based on DOST-PHIVOLCS data
Critical Facilities — OpenStreetMap (OSM)
Administrative Boundaries — NAMRIA
Population Data — Philippine Statistics Authority (PSA), 2020 Census of Population and Housing
0
750 m

GEOGRAPHICAL CONTEXT



- Southern Marikina faces the highest exposure. Low areas near the river and the fault line overlap, making these communities more at risk.

BARANGAY HAZARD EXPOSURE

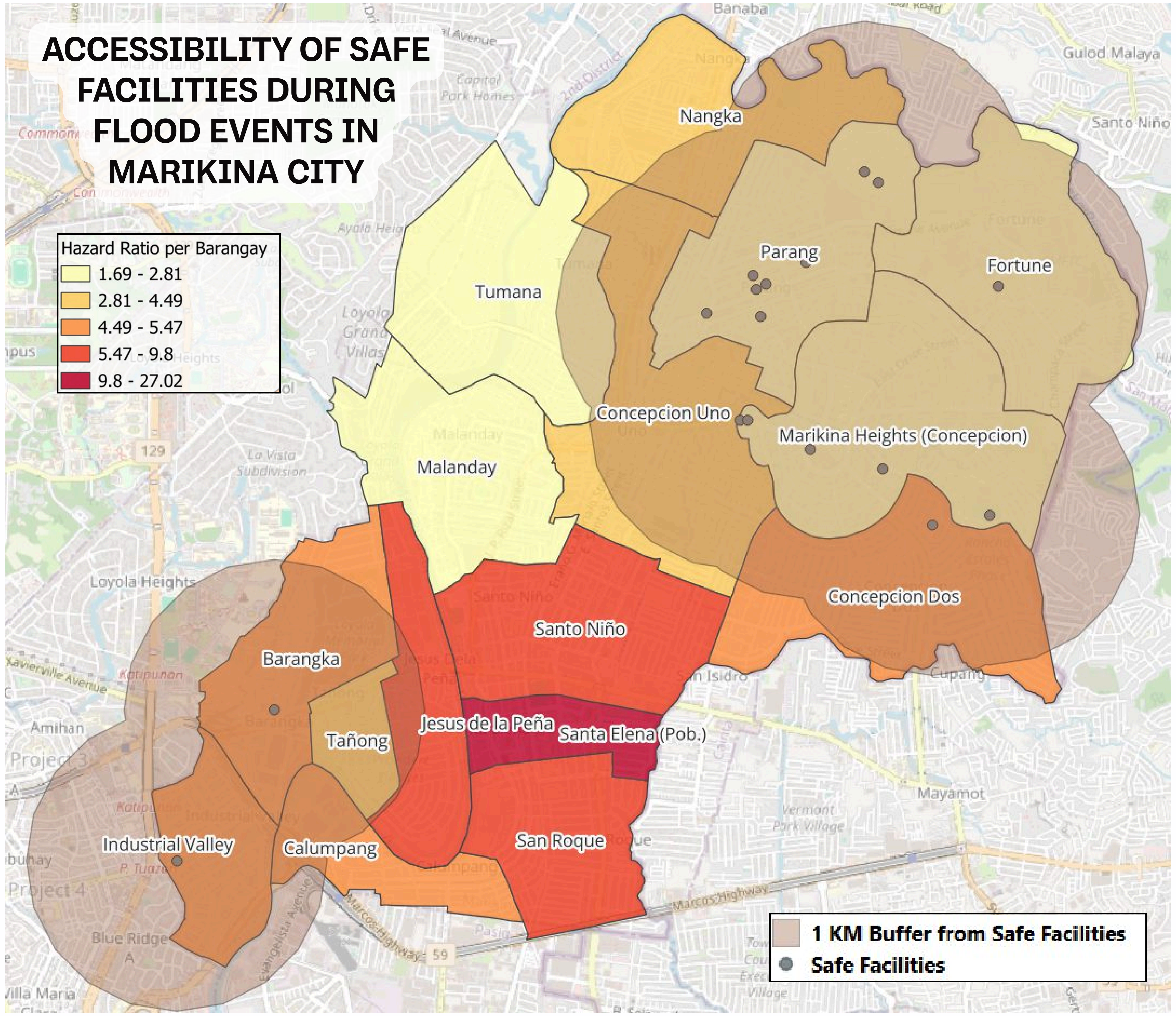
Barangay	Total Facilities	In Flood Zone	Not in Flood Zone	Exposure Score	Population	Hazard Ratio	Class
Santa Elena (Pob.)	10	10	0	20	7,403	27.016	5
Jesus de la Peña	5	5	0	10	10,201	9.803	5
San Roque	8	8	0	16	16,949	9.440	4
Santo Niño	13	13	0	26	28,849	9.012	4
Industrial Valley	5	4	1	9	16,461	5.467	3
Concepcion Dos	8	5	3	13	24,023	5.411	3
Barangka	5	4	1	9	16,639	5.409	3
Calumpang	4	4	0	8	15,602	5.128	3
Tañong	2	2	0	4	8,902	4.493	3
Concepcion Uno	10	10	0	20	44,683	4.476	2
Nangka	8	8	0	16	43,368	3.689	2
Marikina Heights (Concepcion)	9	3	6	12	42,761	2.806	1
Parang	9	1	8	10	40,24	2.485	1
Fortune	5	4	1	9	38,624	2.330	1
Malanday	6	6	0	12	53,886	2.227	1
Tumana	4	4	0	8	47,468	1.685	1

POPULATION PER FACILITY

Barangay	Population	Facilities	People per Facility
Tumana	47,468	4	11,867
Malanday	53,886	6	8,981
Fortune	38,624	5	7,725
Nangka	43,368	8	5,421
Marikina Heights (Concepcion)	42,761	9	4,751
Parang	40,24	9	4,471
Concepcion Uno	44,683	10	4,468
Tañong	8,902	2	4,451

Barangay	Population	Facilities	People per Facility
Calumpang	15,602	4	3,901
Barangka	16,639	5	3,328
Industrial Valley	16,461	5	3,292
Concepcion Dos	24,023	8	3,003
Santo Niño	28,849	13	2,219
San Roque	16,949	8	2,119
Jesus de la Peña	10,201	5	2,04
Santa Elena (Pob.)	7,403	10	740

ACCESSIBILITY



- 1 km buffer around facilities outside flood zones (approx. 10–15 min walking distance)
- Western barangays, particularly Tumana and Malanday, show limited coverage from safe facilities

RESEARCH QUESTIONS

3

Which barangays in Marikina City are most exposed to flood and seismic activity?

Top 4 Most Exposed Barangays (by Hazard Ratio per 10,000 residents):

Barangay	Facilities	Dual-Exposed	Population	Hazard Ratio
Santa Elena (Pob.)	10	10 (100%)	7,403	27.02
Jesus de la Peña	5	5 (100%)	10,201	9.80
San Roque	8	8 (100%)	16,949	9.44
Santo Niño	13	13 (100%)	28,849	9.01

- All four are located in the southern portion of Marikina along the Marikina River floodplain and within the West Valley Fault buffer zone.

Top 3 Most Underserved Barangays (by people per facility):

Barangay	Population	Facilities	People per Facility
Tumana	47,468	4	11,867
Malanday	53,886	6	8,981
Fortune	38,624	5	7,725

- These northern barangays have lower hazard ratios but face facility scarcity, leaving large populations with limited access during emergencies.

FUTURE DIRECTIONS

- Expand study area to include facilities in neighboring cities (Pasig, Antipolo, Quezon City) for a more complete picture of emergency access
- Use road network data and travel-time analysis to model actual accessibility during flood events, including simulating road closures from floodwater
- Assess facility capacity (hospital beds, evacuation floor space, equipment) to determine whether facilities can meet demand during disasters
- Address facility scarcity in underserved northern barangays (Tumana, Malanday, Fortune), potentially through additional infrastructure or mobile emergency units
- Incorporate structural vulnerability data to evaluate earthquake readiness of facilities regardless of location

KEY TAKEAWAYS

Marikina City's critical facilities face significant multi-hazard exposure, with the majority situated in flood hazard zones and within range of the West Valley Fault.

- Southern barangays (Santa Elena, Jesus de la Peña, San Roque, Santo Niño) are the most exposed, located where the river floodplain and fault zone converge
- Northern barangays (Tumana, Malanday, Fortune) are less exposed per capita but face facility scarcity, with up to ~12,000 residents per facility
- Both concentrated hazard exposure and facility access gaps need to be addressed in disaster preparedness planning

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